APPENDIX

SAMPLE DESCRIPTIONS

Sample	Composition (%)				Texture of Sand Fraction		
	Mud	Terrestrial Sand	Marine	Shell & Wood	Grainsize	Sorting	Angularity
1	80	19	0	1	vf-m	P	A
2	10	89	0	1	f-m	M	A
3	100	0	0	0	-	-	-
4	0	100	0	0	m	MW	SA
5	3	97	0	0	f-m	MP	A-SR
6	0	100	0	0	f	MW	S-SR
7	2	96	0	2	m	W	A-SR
8	0	100	0	0	f	MW	SA
9	0	100	0	0	f-vf	VW	A
10	3	97	0	0	f	W	A
11	4	96	0	0	f	MW	A-SA
12	10	90	0	0	f	MW	A-SA
13	40	60	0	0	f-vf	P	A
14	3	92	0	5	f-m	P	A-SR
15	0	90	0	10	f-c	P	SA-SR
16	5	95	0	0	f	W	A-SA
17	18	82	0	0	f	VW	A-SA
18	80	17	0	3	vf	W	A
19	3	96	0	1	f	W	A-SA
20	30	65	0	5	f-vf	MW	A-SA
21	20	75	0	5	f-vf	W	A
22	40	57	0	3	vf	W	A
23	10	18	2	70	f-vf	MW	SA
24	80	13	4	3	f-vf	MP	A-SR
25	10	88	1	1	f-vf	W	A-SA
26	0	18	70	12	f-m	MP	SA-WR
27	0	5	80	15	m	M	R-WR
28	10	70	12	8	vf-m	P	A-SA
29	15	5	70	10	f-m	MP	SR-R
30	0	0	99	1	f-m	M	SR-WR
31	2	7	85	6	f-m	M	SR
32	70	25	0	5	vf	VW	A
33	0	10	74	16	m-c	MP	R-WR
34	50	40	0	10	vf	VW	A
35	98	2	0	0	vf	MW	A
36	98	2	0	0	vf	W	A
37	0	5	93	2	m	MW	SR-R
38	2	48	48	2	m	MW	SA-R
39	5	46	46	3	f-m	M	A-R
40	3	95	0	2	f	M	A-SA
41	5	25	65	5	f-m	MP	SA-R
42	95	3	1	1	vf	W	A
43	50	35	0	15	vf	M	A
44	3	2	70	25	f-m	P	SR-WR
45	0	0	97	3	f-m	W	SR-WR
46	0	0	90	10	f-m	MW	SR-R
47	0	0	100	0	m	MW	SR-R

Sand Textural Symbols

Grainsize		Sorting		Angularity	
vf	very fine	P	poorly	A	angular
f	fine	MP	moderately poorly	SA	subangular
m	medium	M	moderately	SR	subrounded
c	coarse	MW	moderately well	R	rounded
		W	well	WR	well rounded
		VW	very well		

APPENDIX (CONTINUED)

FLOOD SAMPLES

Sample	Composition (%)				Texture of Sand Fraction		
	Mud	Terrestrial Sand	Marine	Shell & Wood	Grainsize	Sorting	Angularity
48	87	8	0	5	f-m	P	A-SA
49	94	3	0	3	f	MW	A-SA
50	25	10	58	7	f-c	P	SA-R
51	90	2	6	2	f-m	P	SA-SR
52	19	44	23	14	f-c	P	A-R
53	77	20	0	3	f-m	P	A-SA
54	98	2	0	0	vf	W	A
55	96	3	0	1	vf	MW	A
56	98	2	0	0	vf	W	A

Sand Textural Symbols

Grainsize			Sorting			Angularity		
vf	-	very fine	P	-	poorly	A	-	angular
f	-	fine	MP	-	moderately poorly	SA	-	subangular
m	-	medium	M	-	moderately	SR	-	subrounded
c	-	coarse	MW	-	moderately well	R	-	rounded
			W	-	well	WR	-	well rounded
			VW	-	very well			

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A Late Devonian Palaeomagnetic Pole for the Mulga Downs Group, Western New South Wales

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ABSTRACT. The thermal stability of the magnetism in specimens from fourteen oriented core samples of red-beds belonging to the Mulga Downs Group was satisfactorily tested using the progressive partial thermal demagnetization technique up to 620°C. Magnetic remanence considered primary and acquired at the time of rock formation was isolated in twenty-one specimens from twelve samples: they yield a palaeomagnetic pole situated at 54°S, 96°E (A_{95} = 11°). This supports an earlier result obtained from the Middle Devonian Housetop Granite of Tasmania but differs substantially from results obtained from the Lochiel Formation of southeastern New South Wales. The result from the Mulga Downs Group is interpreted as emphasizing the discrepancy between the Palaeozoic palaeomagnetic results from the main Australian platform, and those from the Lachlan Fold Belt.

INTRODUCTION

Palaeomagnetic investigations of Early Palaeozoic rock formations have provided us with considerable data on which to base apparent polar wander models for the main platform of the Australian continent. Those data and some interpretations have been reported by McElhinny and Luck (1970), Luck (1972), Embleton (1972a,b), Embleton and Giddings (1974) and McElhinny and Embleton (1974). In addition, results obtained from Siluro-Devonian rock formations sampled in the southeastern region of Australia, reported by Briden (1966) and Luck (1973), have been interpreted within the framework of the plate tectonic hypothesis. McElhinny and Embleton (1974) and Embleton *et al.* (1974), following the earlier interpretations of the tectonic evolution of the Tasman Orogenic Zone by Oversby (1971), Solomon and Griffiths (1972) and Scheibner (1972), postulated that a region in southeastern Australia (the southern part of the Canberra-Molong High) though still rather poorly defined has undergone rigid rotation with respect to the main platform. Serpentinites which outcrop in a 'belt' from Kiandra in the south (the Coolac Serpentinite, Ashley *et al.*, 1971) and northward to Nyngan were regarded as a possible remnant of the Late Palaeozoic (Devonian?) suture. On the plate tectonic model, basic and ultrabasic rocks associated with serpentinites may represent fossil oceanic crust emplaced originally as ophiolite (Coleman, 1971) during a period of either subduction or obduction (Dewey and Bird, 1971). Much of the interpretation was carried out by matching the Siluro-Devonian palaeomagnetic pole positions for the main platform and southeastern regions.

A recent alternative model by Burns and Embleton (1976) explains the discrepancy in the palaeomagnetic data between the main Australian platform and the southeastern region in terms of internal deformation within the Lachlan Fold Belt, without attaching specific significance to the presence of the ultramafic and serpentinite bodies.

The significance of the pole position obtained by Briden (1967) from the Middle Devonian Housetop Granite of Tasmania was uncertain within the plate tectonic scheme. It was originally claimed that the granite had suffered a Tertiary remagnetization, although McElhinny and Embleton (1974) have given an equally plausible explanation of the magnetization being of Devonian age. Support for that pole position now comes from some Upper Devonian sediments from western New South Wales. How these data fit the palaeomagnetic drift scheme remains uncertain, they in fact may emphasize the over-simplicity of the McElhinny-Embleton model. The results do serve to reinforce the principal thesis that the configuration of the Devonian geomagnetic field varies considerably from south-eastern regions to regions further west.

PALAEOMAGNETIC RESULTS

Oriented rock samples were collected using techniques described by Collinson *et al.* (1967, Chap. 1) from three localities in western New South Wales: (i) seven block samples from a section through conglomerates and coarse-grained white/grey sandstones at Mount Manara, about 60 km north of Ivanhoe, (ii) five block samples from a section of white and buff-coloured medium-coarse sandstones at the southern end of the Manara Hills, about 15 km southwest of Mount Manara, and (iii) fourteen core samples collected with a portable hand drill from a 20-30 m escarpment outcrop about 8 km northeast of Milton Grove Station (latitude 32.8°S, longitude 143.5°E). Only samples from locality (iii) retained measureable stable components of magnetization. None of the samples from localities (i) and (ii) produced repeatable results upon partial thermal demagnetization.

The beds sampled near Milton Grove Station trend approximately east-northeasterly at 65° and dip to the northwest at 30°. This sequence of red sediments belongs to the Mulga Downs Group, generally regarded as Late Devonian in age. Roberts *et al.* (1972) cite evidence from Ritchie (1969) which supports a Middle Devonian age for the lower part of the succession. The highest beds may be as young as Early Carboniferous.

Two specimens were cut from each core and



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